

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T41 · Bearings, Scale Drawing and Constructions

With Answers

Questions with full worked solutions

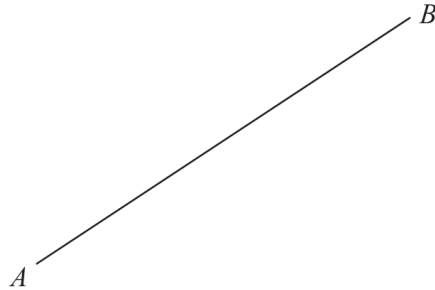
■ 26 pages

FAMILY MAP

Bearings Scale Drawing Constructions

F01	Straightedge-and-compass construction	5 Q 10 marks
F02	Bearing geometry with trigonometry	3 Q 16 marks
F03	Scale drawing and map location	2 Q 4 marks
F04	Direct bearing language	1 Q 2 marks

- 4 Using ruler and compasses only, construct the perpendicular bisector of the line AB .
Show all your construction lines.



(Total for Question 4 is 2 marks)

Worked Solution 19

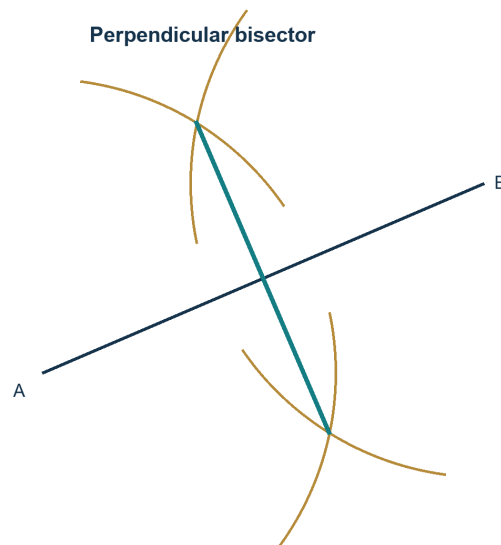
Perpendicular bisector construction

UNIT 2
2 marks**Construct arcs**

With compass radius greater than half AB , draw intersecting arcs from A and B on both sides.

Draw the bisector

Join the two arc intersections; this line is perpendicular to AB and bisects it.

**Final answer**

A correct perpendicular bisector with two pairs of construction arcs

Question 14

4MA1-1H Q18 demand

ORIGINAL QUESTION UNIT 2

5 marks

- 18 The diagram shows the positions of three villages, A , B and C

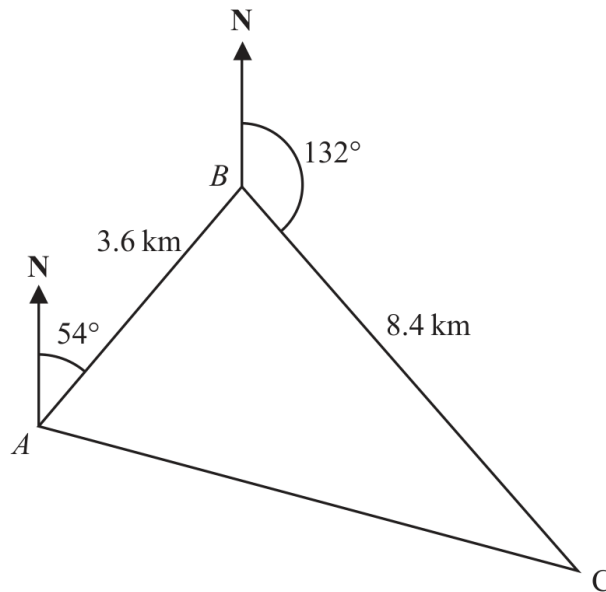


Diagram **NOT**
accurately drawn

The bearing of B from A is 054°

The bearing of C from B is 132°

Melur walks from A to B

She then walks from B to C and from C to A

Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.

Give your answer in hours and minutes.

..... hours minutes

Worked solution 14

Bearings and total walking time

MODULAR UNIT 2

5 marks

Demand – Bearings and total walking time

Find angle ABC

$$\angle ABC = 54^\circ + (180^\circ - 132^\circ) = 102^\circ.$$

Find AC

$$AC^2 = 3.6^2 + 8.4^2 - 2(3.6)(8.4) \cos 102^\circ \Rightarrow AC \approx 9.802 \text{ km.}$$

Find time

$$t = \frac{3.6 + 8.4 + 9.802}{6} = 3.6337 \text{ h} = 3 \text{ h } 38 \text{ min (approximately).}$$

Final answer

3 hours 38 minutes

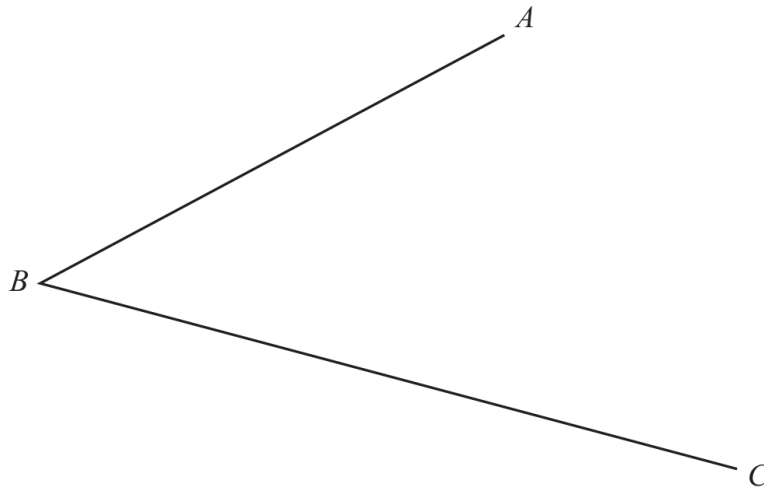
Original question 03

4MA1-1H Q2 M01

MODULAR UNIT 2

2 marks

- 2 Use ruler and compasses only to construct the bisector of angle ABC .
You must show all your construction lines.



Worked solution 03

Construct an angle bisector

MODULAR UNIT 2

2 marks

M01 – Construct an angle bisector**Make the first arc**

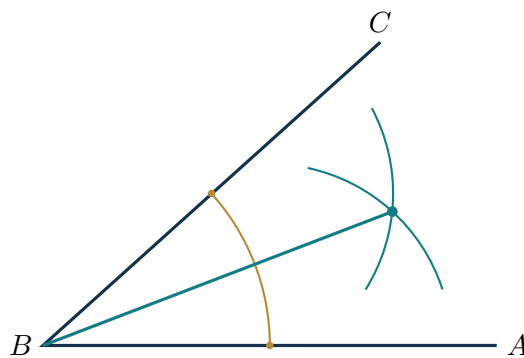
With centre B, draw an arc cutting BA and BC at two points.

Make equal arcs and join

With the same radius draw arcs from those two points; join B to their intersection. This line bisects $\angle ABC$.

Final answer

The constructed ray from B that bisects angle ABC, with all compass arcs shown

Completed solution diagram: Angle bisector with construction arcs

Original question 35

4MA1-2H Q24 M01

MODULAR UNIT 2

6 marks

24 A , B and C are three points on horizontal ground.

$$AB = 8.4 \text{ metres} \quad BC = 9.2 \text{ metres}$$

B is on a bearing of 067° from A

C is on a bearing of 129° from B

Calculate the bearing of A from C

Give your answer correct to the nearest degree.

Original question 35 (continued)

4MA1-2H Q24 M01

MODULAR UNIT 2

6 marks

.....

Worked solution 35

Find a bearing using the cosine and sine rules

MODULAR UNIT 2

6 marks

M01 – Find a bearing using the cosine and sine rules**Find the included angle**

$$\angle ABC = 67^\circ + (180^\circ - 129^\circ) = 118^\circ.$$

Find AC

$$AC^2 = 8.4^2 + 9.2^2 - 2(8.4)(9.2) \cos 118^\circ \Rightarrow AC = 15.09 \dots$$

Find an angle

Use the sine rule to obtain the angle at C, about 29–30°.

Convert to a bearing

The bearing of A from C is 280° (nearest degree).

Final answer

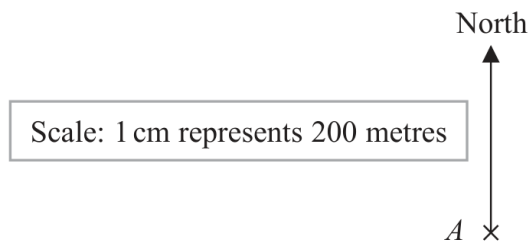
280 degrees

Original question 18

4MA1-2H Q5 Q5(a)

ORIGINAL QUESTION**3 marks**

- 5 The scale diagram shows the position on a map of a house, A



House C is on a bearing of 110° from A

The distance from A to C is 700 m

- (a) Mark the position of C on the diagram with a cross (×)
Label your cross C

(3)

Worked solution 18

Chapter: Bearings, Scale Drawing & Constructions

MODULAR UNIT 2

3 marks

Q5(a) – Chapter: Bearings, Scale Drawing & Constructions

Family: Scale drawing and map location • Variant: Plot a point from bearing and distance

Method700 m corresponds to $700/200=3.5$ cm on the map.*Why: Use the scale 1 cm represents 200 m.***Method**From A, draw a ray on a bearing of 110° clockwise from north.*Why: Bearings are measured clockwise from north.***Method**

Mark and label C at the end of this 3.5 cm ray.

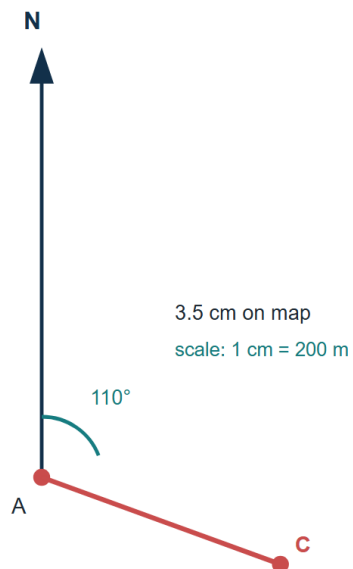
*Why: The distance and bearing locate C uniquely.***Final answer****map position:** C at 3.5 cm on bearing 110° from A

Completed diagram 18

Chapter: Bearings, Scale Drawing & Constructions

MODULAR UNIT 2

3 marks

Q5 • Completed bearing mapC is 700 m from A on a bearing of 110° 

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Original question 19

4MA1-2H Q5 Q5(b)

ORIGINAL QUESTION**1 marks**

(b) Write the scale of the map in the form $1:n$

1 :
(1)

Worked solution 19

Chapter: Bearings, Scale Drawing & Constructions

MODULAR UNIT 2

1 marks

Q5(b) – Chapter: Bearings, Scale Drawing & Constructions

Family: Scale drawing and map location • Variant: Convert a map scale to 1:n

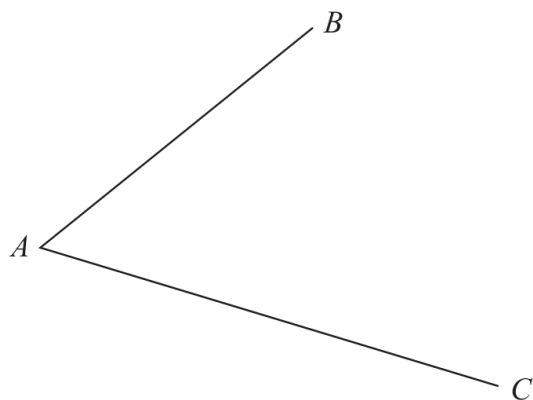
Method

1 cm represents 200 m, so 1 cm represents 20000 cm; the scale is 1:20000.

*Why: Convert both units to centimetres.***Final answer**

scale: 1:20000

- 5 Use ruler and compasses to construct the bisector of angle BAC .
You must show all your construction lines.



Worked solution

January 2020 | Question 5 | 2 marks

Family: Straightedge-and-compass construction • Variant: Bisector of an angle

Method / working

1. With centre A, draw an arc cutting AB and AC.
2. From the two cut points draw equal arcs; join A to their intersection.

Final answer: Construct the perpendicular bisector ray of angle BAC

16 The diagram shows the positions of three ships, A , B and C .

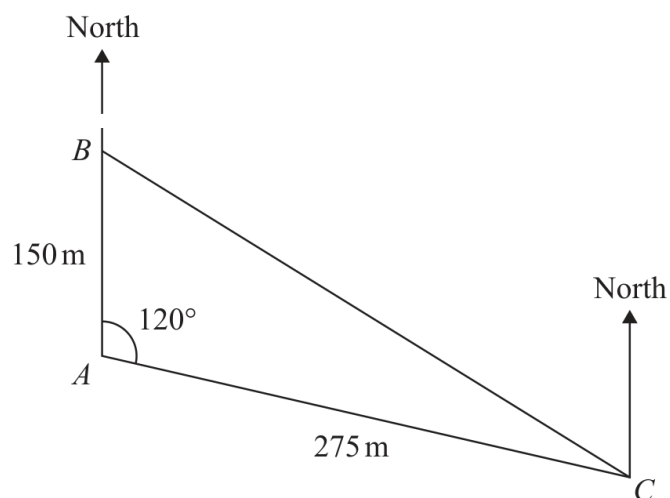


Diagram **NOT**
accurately drawn

Ship B is due north of ship A .

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B .

Give your answer correct to the nearest degree.

○

Worked solution

January 2021 | Question 16 | 5 marks

Family: Bearing geometry with trigonometry | Variant: Sine or cosine rule supports a final bearing

METHOD / WORKING

1. Use the cosine rule: $BC^2 = 150^2 + 275^2 - 2(150)(275)\cos 120^\circ$.

2. $BC = 373.4 \dots$ m.

3. Use the sine rule to find the angle at A or C.

4. The required bearing from B is $120^\circ + 20.3^\circ \sim 140.3^\circ$.

5. To the nearest degree, the bearing is 140° .

FINAL ANSWER

140°

June 2021 | 4MA1-2H | Question 3 | 2 marks

Chapter OL-T41: Bearings, Scale Drawing & Constructions

- 3 The bearing of Paris from London is 149°
Work out the bearing of London from Paris.

o

Worked solution

June 2021 | Question 3 | 2 marks

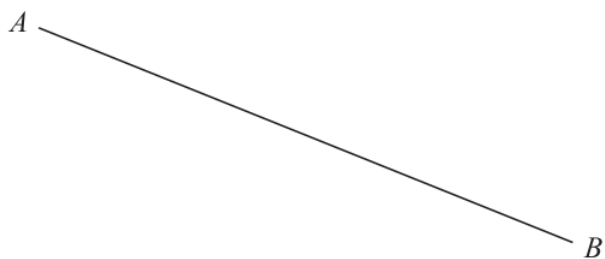
Family: Direct bearing language • Variant: Reverse bearing

Method / working

1. Reverse the bearing by adding 180° to 149° .
2. 329° is already in the required 0° - 360° range.

Final answer: 329°

- 3 Use ruler and compasses only to construct the perpendicular bisector of line AB
You must show all your construction lines.



(Total for Question 3 is 2 marks)

Answer reference | January 2023 | Question 3

Family: Straightedge-and-compass construction • Variant: Perpendicular bisector of a segment

3	Two pairs of intersecting arcs with equal radii centre <i>A</i> and <i>B</i>		2	M1 for arcs that intersect within or on the guidelines or correct perpendicular bisector without arcs.
	<i>Working required</i>	Bisector with construction arcs		A1 for a fully correct bisector with two intersecting arcs
				Total 2 marks

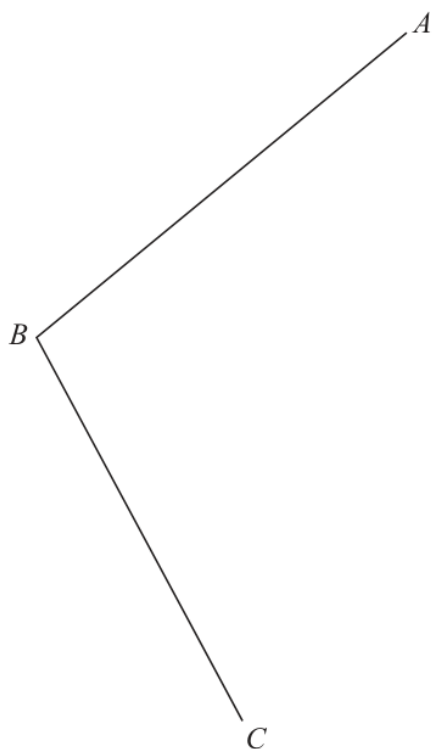
Worked solution

January 2023 | Question 3 | 2 marks

Family: Straightedge-and-compass construction • Variant: Perpendicular bisector of a segment

Official final mark-scheme pages are embedded immediately after this question.

- 3 Using ruler and compasses only, construct the bisector of angle ABC
Show all your construction lines.



(Total for Question 3 is 2 marks)

Worked solution

November 2025 | Question 3 | 2 marks

Family: Straightedge-and-compass construction • Variant: Construct a triangle or other 2D shape

Construction answer

To construct the bisector of angle ABC:

1. Use centre B to draw an arc crossing BA and BC.
2. Use the same compass width from the two crossing points to make two arcs inside the angle.
3. Join B to the point where these two arcs meet.

Answer: That line is the required angle bisector.